

Bytes vs Bytearray vs List

Common Features

Feature	Description
Indexing	Supports both positive and negative indexing
Data Extraction	Elements can be accessed using index positions
Insertion Order	Order of insertion is preserved
Iteration	Can be traversed using loops (for loop, etc.)

Uncommon Features

Feature	bytes	bytearray	list
Mutability	■ Immutable	■ Mutable	■ Mutable
Data Type Restriction	Homogeneous (0–255 only)	Homogeneous (0–255 only)	Heterogeneous (any data type)
Value Range	0–255 only	0–255 only	Any valid Python object
Dynamic Size	■ Fixed once created	■ Can grow/shrink	■ Fully dynamic

Dynamic Behavior

Type	Behavior
bytes / bytearray	Values restricted to integers (0–255), used for binary data
list	Fully dynamic — can store any object and grow/shrink freely
tuple (for comparison)	Immutable, not dynamic

Use Cases

Type	Real-World Usage
bytes	Handling binary files, network data, images
bytearray	When binary data needs modification before writing/sending
list	General-purpose data handling in programs, mixed-type collections